

DIGITAL IMAGE PROCESSING R2026

SUBMITTED TO: DR. JAMEEL AHMAD

HOMEWORK ASSIGNMENT # 1

MUHAMMAD UMER: F2023266912

HASEEB ULLAH: F20232661009

These are the written theory questions, for coding questions, see .ipynb files

Q # 1(a) Numericals

M.Umer 912
Haseeb 1009

(i) $16 \text{ pixels} \times 8 \text{ bits/pixel} = 128 \text{ bits}$

(ii) $16 \text{ pixels} \times 3 \text{ bits/pixel} = 48 \text{ bits}$

Bytes = $48 \text{ bits} / 8 = 6 \text{ bytes}$

Compression Ratio = $\frac{128 \text{ bits}}{48 \text{ bits}} = 2.67$

(iii) Total file size =

$54 \text{ bytes (header)} + 16 \text{ bytes (pixel data)}$

$= 70 \text{ bytes}$

(iv) By default cv2.imread loads images in the BGR mode (even if the source is grayscale) for consistency across OpenCV's pipelines, which mostly assumes 3-channel color images - use the flag cv2.IMREAD_GRAYSCALE to load it as a true single-channel array.

Q#2 (a) Numerical

M. Umer 912
Haseeb 1009

$$A = \begin{bmatrix} 250 & 10 & 90 \\ 60 & 200 & 30 \\ 15 & 5 & 180 \end{bmatrix} \quad B = \begin{bmatrix} 20 & 250 & 40 \\ 80 & 90 & 220 \\ 10 & 240 & 90 \end{bmatrix}$$

$$(i) \quad A+B \Rightarrow \text{mod}(256) \Rightarrow \begin{bmatrix} 14 & 4 & 130 \\ 140 & 34 & 250 \\ 25 & 245 & 14 \end{bmatrix}$$

$$(ii) \quad \min(a+b, 255) \Rightarrow \begin{bmatrix} 255 & 255 & 130 \\ 140 & 255 & 250 \\ 25 & 245 & 255 \end{bmatrix}$$

$$(iii) \quad \max(a-b, 0) \Rightarrow \begin{bmatrix} 230 & 0 & 50 \\ 0 & 110 & 0 \\ 5 & 0 & 90 \end{bmatrix}$$

(iv) Saturation is preferred because it preserves visual and physical plausibility; when a pixel value overflows, it clamps to the nearest valid extreme (255 or 0), rather than wrapping to an unrelated often opposite value.

Q # 3 (a) Numericals

M/Mer 912
Hoseeb 1009

(i) $f_{d_2}(m) = [10, 250, 30, 200]$

(ii) $f_{d_4}(m) = [10, 30]$

(iii) Us sample

$$m = [10, 250, 30, 200]$$

$$\hat{f}(m) = [10, 10, 250, 250, 30, 30, 200, 200]$$

original

$$[10 \quad 100 \quad 250 \quad 40 \quad 30 \quad 220 \quad 200 \quad 15]$$

Reconstructed

$$[10 \quad 10 \quad 250 \quad 250 \quad 30 \quad 30 \quad 200 \quad 200]$$

$$MSE = \frac{\text{Squared sum}}{\text{to no.}} = \frac{122525}{8} = 15315.6$$

(iv) Aliasing occurs when high frequency signal changes are sampled too sparsely to be captured correctly, and they instead masquerade as false, lower frequency patterns as seen here, where disregarding the wildly swinging odd samples (100, 40, 220, 15) without pre filtering caused the reconstructed sequence to completely miss those swings producing large MSE.

Q# 4(a) Numericals

M. Umer 912
Haseeb 1009

$$x = [18, 63, 97, 130, 175, 210, 240, 5]$$

(i) $L = 4$ (2 bits/pixel)

$$\Delta = 256/L \Rightarrow 256/4 \Rightarrow 64$$

$$q(x) = \lfloor x/\Delta \rfloor \cdot \Delta + \Delta/2$$

$$q(x) = [32, 32, 96, 160, 160, 224, 224, 32]$$

(ii) $L = 2$ (1 bit/pixel)

$$q(x) = [64, 64, 64, 192, 192, 192, 192, 64]$$

(iii) Error = $[x - q(x)]$

$$\text{or } q(x) = [-14, 13, 1, -30, 15, -14, 16, -27]$$

$$MSE = \frac{1}{N} \sum_{i=1}^N e(x_i)^2$$

$$MSE = 433$$

- (iv) As L increases, Δ (step size) decreases,
so MSE decreases (fewer, smaller, rounding errors)
As L decreases, MSE increases.

Artifact: Very few levels cause false contouring/
posteriorization, smooth gradients turn into visible
flat bands instead of a smooth transition.

Q#5 (a) Numericals.

M.Umer 912
Haseeb 1009

(i) Generate a 5x5 and pass into the gaussian function

-2	0.0183	-0.0821	-0.1353	0.0821	0.0183
1	0.0821	-0.3679	0.6065	-0.3679	0.0821
0	0.1353	0.6065	1	-0.6065	-0.1353
1	-0.0821	-0.3679	0.6065	-0.3679	0.0821
2	0.0183	-0.0821	-0.1353	0.0821	0.0183
	-2	-1	0	1	2

(ii) Normalized values ~~5x5~~ (Sum = 1)

0.0030	0.0133	0.0219	0.0133	0.0030
0.0133	0.0596	0.0983	0.0596	0.0133
0.0219	0.0983	0.1621	0.0983	0.0219
0.0133	0.0596	0.0983	0.0596	0.0133
0.0030	0.0133	0.0219	0.0133	0.0030

(iii) Row sums are $\Rightarrow$ 0.0545, 0.2441, 0.4025,
0.2441, 0.0545

Sum $\Rightarrow$ 1

(iv) Normalization keeps the kernel weights to 1 so convolution computes a weighted average of neighbouring pixels rather than scaling intensities up or down. This preserves the image's overall brightness while smoothing/blurring it.